

Solution of Quiz-02
GPE201 (Earth and Planetary System)

Date of Exam.: 10 April 2024 Time of Exam.: 19:15 – 19:45 hr Maximum marks: 10

SOLUTION

Q.1

A climate feedback refers to a process that amplifies or dampens the effects of climate forcings. In other words, feedbacks can either reinforce (positive feedback) or counteract (negative feedback) the initial forcing, leading to either further warming or cooling of the climate system.

An example of a positive feedback is the ice-albedo feedback. As temperatures rise, ice and snow melt, reducing the Earth's surface albedo (reflectivity). This decrease in albedo means that more sunlight is absorbed by the Earth's surface, leading to further warming and more ice melt. This cycle continues, amplifying the initial warming.

An example of a negative feedback is the water vapor feedback. As the atmosphere warms, the amount of water vapor it can hold increases. Since water vapor is a greenhouse gas, this leads to additional warming. However, water vapor also forms clouds, which can reflect sunlight back into space, thereby cooling the Earth's surface. This cloud formation acts as a negative feedback, counteracting some of the initial warming caused by increased water vapor.

Q.2

Three key factors, besides insolation and greenhouse gases, that have played a significant role in altering Earth's climate over geological time are;

- (i) plate movement,
- (ii) earth's orbital changes, and
- (iii) volcanism

Q.3

As per Wien's first law (Wien's Displacement Law), the wavelength $\lambda_{\max}$ at an isotherm peak shifts toward short wavelengths when temperature T increases. According to this law:

$$\lambda_{\max} T = \sigma_w$$

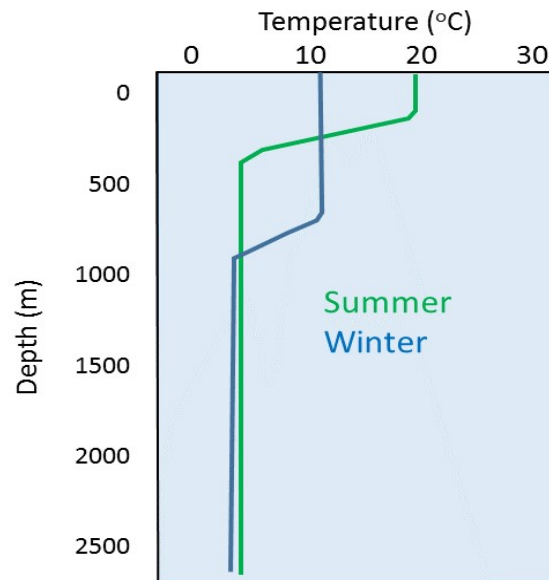
Where, $\sigma_w = 2.898 \times 10^{-3}$ mK which is Wien's displacement constant.

Therefore,

$$\begin{aligned} T &= \sigma_w / \lambda_{\max} \\ &= (2.898 \times 10^{-3}) / (400 \times 10^{-9}) \\ &= 7245 \text{ K} \\ &= (7245 - 273)^\circ\text{C} \\ &= 6972^\circ\text{C} \end{aligned}$$

Q.4

Vertical temperature distribution in the ocean during summer and winter are presented below:



Q.5

Climate sensitivity refers to the equilibrium global surface temperature change that would occur in response to a doubling of atmospheric carbon dioxide (CO_2) concentrations.

- The 2013 Intergovernmental Panel on Climate Change (IPCC) fifth assessment report suggests that ECS has a 'likely' range of 1.5 - 4.5°C. It adds ECS is extremely unlikely to be below 1°C and very unlikely greater than 6°C. The wide range accounts for all evidence.
- The IPCC estimate TCR has a likely range of 1.0 - 2.5°C and is extremely unlikely more than 3°C.
- The 2021 Intergovernmental Panel on Climate Change (IPCC) report suggests that the equilibrium climate sensitivity (ECS) is likely in the range of 2.5°C to 4.0°C, with a best estimate of about 3.0°C. This means that if atmospheric CO_2 concentrations were to double from pre-industrial levels (which were around 280 parts per million), global average temperatures would eventually rise by about 2.5°C to 4.0°C, with a central estimate of 3.0°C.

Q.6

The Rayleigh-Jeans catastrophe, also known as the ultraviolet catastrophe, was a problem in classical physics that arose in the late 19th century when attempting to describe the spectral distribution of blackbody radiation. According to classical electromagnetic theory, the intensity of radiation emitted by a blackbody should increase without limit as the frequency of the radiation approaches infinity. This prediction, however, contradicted experimental observations, leading to the so-called "catastrophe."

The Rayleigh-Jeans catastrophe played a crucial role in the development of quantum mechanics by highlighting the failure of classical physics to explain certain phenomena, specifically the blackbody radiation spectrum. To resolve the issue of the Rayleigh-Jeans catastrophe, Max Planck proposed a revolutionary idea in 1900. He suggested that electromagnetic radiation can be quantized, meaning it can only exist in discrete packets of energy, or quanta. This assumption was a radical departure from classical physics, where energy was thought to be continuous. Planck introduced a constant, now known as Planck's constant (h), into his equations to describe this quantization of energy.

Albert Einstein further advanced this concept in 1905 with his explanation of the photoelectric effect, which demonstrated that light behaves as both a wave and a particle (later termed a photon). These early quantum ideas laid the foundation for modern quantum mechanics, which revolutionized our understanding of physics at the atomic and subatomic levels.

Q.7

The four primary gases in Earth's atmosphere are nitrogen, oxygen, argon, and carbon dioxide. Their relative proportions are approximately:

Nitrogen (N₂): about 78.08%

Oxygen (O₂): about 20.95%

Argon (Ar): about 0.93%

Carbon dioxide (CO₂): about 0.04%

Q.8

AOD stands for Aerosol Optical Depth, which is a measure of the attenuation of sunlight due to aerosol particles in the atmosphere. It quantifies how much the aerosols in the atmosphere reduce the transmission of light. AOD is a unitless quantity typically ranging from 0 to 1, but can be higher under certain conditions.

For extremely clean atmospheric conditions, AOD values can be very low, close to 0. For example, over remote oceans or pristine regions with minimal human activity and aerosol emissions, AOD values can be as low as 0.01 to 0.05.

In contrast, for very hazy conditions with high aerosol concentrations, AOD values can be about 0.45 and some time much higher, exceeding 1. These high AOD values are often observed in urban areas or regions affected by pollution, dust storms, or biomass burning.

Q.9

The distribution of temperature in ocean water is influenced by several factors, including:

- i. Latitude
- ii. Inequality in the distribution of water and land
- iii. Prevailing wind
- iv. Oceanic currents
- v. Distance of the earth from the sun
- vi. Underwater topography
- vii. Local weather conditions
- viii. Location and shape of sea
- ix. Duration of day
- x. Number of sunspots activities

Q.10

As per Wien's second law, the maximum monochromatic exitance, $M_{\lambda_{\max}}^0$, is proportional to the fifth power of temperature T , which is:

$$\begin{aligned} M_{\lambda_{\max}}^0 &= BT^5 \\ &= 1.28 \times 10^{-11} \times (2150)^5 \\ &= 5.88 \times 10^5 \text{ Wm}^{-2}\mu\text{m}^{-1} \end{aligned}$$